# utility10_08-tap-litre-meter

# Tap Litre Meter: manufacturing manual

> **Status: design only.** Nothing here has been built or tested as hardware. This manual tells you exactly what to make, buy, assemble, program and test, and how to accept or reject a unit; expect to correct it after the first prototype run (5 units).

**Product summary.** An inline flow sensor on a tap or hose; the display shows litres this session and today, and a buzzer rings when a set target (e.g. 20 L bucket) is reached.

**Product line / price in the plan:** Tier B · low-voltage electronics · prototype Rs 1200 · price Rs 1199. Status: designed and verified as far as a computer can; no prototype built, nothing field-tested.

## 1. What you are building: deliverables

| Deliverable | Where | State |
|---|---|---|
| PCB: Gerbers, BOM, CPL, KiCad project | `utility10/pcb/release/tap/` | **order-ready: every automatic check passes** (DRC errors 0, netlist match True) |
| Firmware (`boards/tap_litre.hex`) | `utility10/firmware/release/8MHz/boards/tap_litre.hex`, source `utility10/firmware/src/main_board.cpp` (seeti: `seeti_prod.cpp`), drivers `board_drivers.h`, engine `utility10/firmware/lib/utilcore.c` | engine host-tested (115 checks); board firmware compiles but is **not yet run on hardware** (see section 6) |
| Enclosure STL (prototype shell) | `utility10/08-tap-litre-meter/enclosure_body.stl`, `enclosure_lid.stl` and the board's `4_enclosure/` | prototype shell only |
| Mechanical / bought-in parts | section 3 | listed |
| Test sheet | `manuals/templates/eol_test_sheet.md` | template |

## 2. Before you start

- Read `manuals/STANDARD_PROCEDURES.md` (sections A-F, J, K).
- Tools: soldering iron, flux, multimeter, USBasp programmer + avrdude, calipers, 3D printer or print service, bench supply with current limit.
- Power source for this product: **3 x AAA**.

## 3. Bill of materials

### 3a. Circuit board parts (25 parts, 3 test points)

SMD parts marked `SMT (JLC)` are assembled by the fab from the JLC BOM; parts marked `hand/THT` are soldered by you. Every LCSC number was checked against an LCSC/JLCPCB listing on 2026-11-01; re-check stock on the day you order.

| Ref | Value | Package | Part (manufacturer) | LCSC | Assembly |
|---|---|---|---|---|---|
| U1 | ATmega328P-AU | TQFP-32_7x7mm_P0.8mm | ATMEGA328P-AU | C14877 | SMT (JLC) |
| C1 | 100n | C_0603_1608Metric | 0603 100nF X7R | C14663 | SMT (JLC) |
| C2 | 100n | C_0603_1608Metric | 0603 100nF X7R | C14663 | SMT (JLC) |
| C3 | 100n | C_0603_1608Metric | 0603 100nF X7R | C14663 | SMT (JLC) |
| R7 | 10k | R_0603_1608Metric | 0603 10k 1% | C25804 | SMT (JLC) |
| J2 | ISP | PinHeader_2x03_P2.54mm_Vertical | ZHOURI 2.54-2*3 | C5116479 | hand/THT |
| J1 | BATT 3xAA | JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical | JST B2B-PH-K-S(LF)(SN) | C131337 | hand/THT |
| SW1 | Power | SW_SPDT_PCM12 | C&K PCM12SMTR | C221841 | SMT (JLC) |
| Q2 | AO3401A | SOT-23 | AO3401A | C15127 | SMT (JLC) |
| C4 | 10u | C_0805_2012Metric | 0805 10uF X5R | C15850 | SMT (JLC) |
| BZ1 | Buzzer | Buzzer_12x9.5RM7.6 | YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz) | C781869 | hand/THT |
| Q1 | MMBT3904 | SOT-23 | MMBT3904 | C20526 | SMT (JLC) |
| D2 | 1N4148WS | D_SOD-323 | 1N4148WS | C2128 | SMT (JLC) |
| R6 | 1k | R_0603_1608Metric | 0603 1k 1% | C21190 | SMT (JLC) |
| D1 | Green | LED_0603_1608Metric | Everlight 19-217/GHC-YR1S2/3T (LCSC showed it out of stock on the search day: check) | C72043 | SMT (JLC) |
| R5 | 1k | R_0603_1608Metric | 0603 1k 1% | C21190 | SMT (JLC) |
| SW2 | Reset / target | SW_SPST_B3U-1000P | Omron B3U-1000P | C231329 | SMT (JLC) |
| J3 | Flow sensor | JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical | JST B3B-PH-K-S(LF)(SN) | C131339 | hand/THT |
| R8 | 1k | R_0603_1608Metric | 0603 1k 1% | C21190 | SMT (JLC) |
| R9 | 10k | R_0603_1608Metric | 0603 10k 1% | C25804 | SMT (JLC) |
| C9 | 100n | C_0603_1608Metric | 0603 100nF X7R | C14663 | SMT (JLC) |

| Ref | Value | Package | Part (manufacturer) | LCSC | Assembly |
|---|---|---|---|---|---|
| D4 | TVS 5V bidir | D_SOD-323 | Nexperia PESD5V0S1BA,115 | C19224 | SMT (JLC) |
| J4 | OLED 0.91in I2C | JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical | JST B4B-PH-K-S(LF)(SN) | C131334 | hand/THT |
| R10 | 4.7k | R_0603_1608Metric | 0603 4.7k 1% | C23162 | SMT (JLC) |
| R11 | 4.7k | R_0603_1608Metric | 0603 4.7k 1% | C23162 | SMT (JLC) |

### 3b. Bought-in and mechanical items (estimates for about 2,000 units; replace with quotes)

| Item | Rs |
|---|---|
| Hall flow sensor G1/2 | 90 |
| MCU | 45 |
| Display | 35 |
| Battery | 25 |
| PCB | 30 |
| Enclosure | 50 |
| Hose adapters | 35 |
| Packaging | 30 |
| Assembly + test | 40 |

## 4. Step 1: fabricate the PCB

Upload `utility10/pcb/release/tap/1_fabrication/tap_litre_meter_gerbers.zip` with the settings in STANDARD_PROCEDURES section A. Finish: HASL lead-free. Board size and layer list are in `utility10/pcb/release/tap/README.md`.

## 5. Step 2: assemble the board

1. Order SMT assembly as in STANDARD_PROCEDURES section B (or hand-solder the SMD parts in section D order).
2. Solder the through-hole parts listed `hand/THT` in 3a.
3. Clean, inspect under magnification, and compare with `2_assembly/*_assembly_top.pdf`.

## 6. Step 3: program the microcontroller

1. Connect a USBasp to the 6-pin ISP header J2 (pin 1 = square pad) with the board powered (3 x AAA).
2. Set fuses: `avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m` (internal 8 MHz RC, CKDIV8 off, brown-out 2.7 V (the 8 MHz speed-grade limit): the unit resets cleanly when the battery is flat).
3. Flash and verify: `avrdude -p m328p -c usbasp -U flash:w:utility10/firmware/release/8MHz/boards/tap_litre.hex:i` then `-U flash:v:utility10/firmware/release/8MHz/boards/tap_litre.hex:i`.
4. Remove the programmer, power-cycle the board, and confirm the start-up behaviour (section 9).

If you change the supply type (for example USB instead of batteries) use the hex built for that clock (`4MHz` for 2 x AAA, `8MHz` for every other tool).

Firmware pin map (generated from the PCB spec, so firmware and board agree):

| Arduino pin | ATmega pin | Net |
|---|---|---|
| D3 | 1 | FLOW |
| D7 | 11 | LED |
| D9 | 13 | BUZ |
| A4 | 27 | SDA |
| A5 | 28 | SCL |
| D2 | 32 | BTN |

**What is verified:** the control logic (`utilcore.c`) passes 115 host tests. **What is not:** the display, OLED, RTC and HX711 drivers and the button flows in `main_board.cpp` were written from datasheets and only compile-checked. Budget bring-up time for them on the first prototype.

## 7. Step 4: electrical bring-up

1. Blow/spin the sensor: FLOW (TP3) pulses; firmware counts.
2. Run 10 L from a calibrated bucket: adjust pulses-per-litre.

## 8. Step 5: sensors, cables and mechanical assembly

- **Flow sensor.** Use a flow sensor with a lot calibration sheet (a Hall-effect turbine type, 3 wires: +, GND, signal into J3). Connect with hose barbs/G1/2 fittings; use thread tape. Pot or conformally coat the electronics module and keep the flow sensor, not the electronics, in the water path.

## 9. Step 6: enclosure and final assembly

1. Print the shell (STANDARD_PROCEDURES section G). Enclosure: `enclosure_body.stl` / `enclosure_lid.stl` (70×50×24 mm, 2.4 mm walls, watertight): a simple prototype shell; production tooling needs a proper design.
2. Place the board on the standoffs (4 mm), fasten with M2.5 screws (6 mm), route cables through the cut-outs, close the lid.
3. Label: serial number, rev, battery type/polarity, warnings.

## 10. Step 7: calibration

- Pulses per litre (450 is a typical datasheet value) must be calibrated per sensor lot: run 10 L from a calibrated bucket, divide pulses by 10, enter in the setup (stored in EEPROM).

## 11. Step 8: end-of-line functional test (every unit)

1. Run 10 L from a calibrated bucket: reading 10 L +-0.2 L (adjust pulses/L).
2. Stop flow for 5 s: the session ends; start again: a new session starts; set the target to 10 L: the buzzer rings at 10 L.

Record results on `manuals/templates/eol_test_sheet.md`. Accept only if every line passes.

## 12. Packaging, documents, compliance

- In the box: unit, batteries or note on the required adapter, mounting accessories, quick-start card (`manuals/templates/quick_start_card.md`), warranty card.
- Compliance items to verify before sale: Battery safety, BIS requirement for any bought-in adapter, EMC, e-waste EPR, Legal Metrology labelling; no safety claims..
- Labelling: STANDARD_PROCEDURES K.

## 13. Troubleshooting (product specific)

| Symptom | Cause / action |
|---|---|
| Reads high/low | calibration not done or flow rate outside the sensor range |
| Leaks | re-tape threads; never over-tighten plastic fittings |

General issues: STANDARD_PROCEDURES section L.

## 14. Known limits and risks

Host tests: litres from pulses, a session ends after 5 s without flow, target alarm at 10 L, new session starts cleanly, day total adds. Pulses per litre (450) is a typical datasheet value, **the pack includes a 10 L bucket calibration**.

Accuracy at low flow; sensors clog with sediment (add a filter screen); seals.

Nothing in this manual replaces testing with real users, a design review by a PCB engineer, and the compliance checks above.